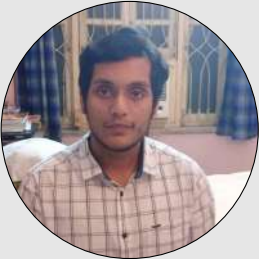


NAME

Maitreya Gupta



Personal

DOB: 02.10.2000  
Hobbies: Karate, KickBoxing,  
Game Dev, Coding  
Languages: Hindi, English,  
Bengali

Technical Skills

Programming:  
Python, C++, Java, C  
Web:  
React, Flask, Django,  
HTML/CSS  
Robotics:  
ROS, ESP32, MediaPipe, YOLO  
Tools:  
Git, VS Code, Godot, Three.js

Social Links

- GitHub
- HackerRank
- TryHackMe
- LinkedIn
- Portfolio Website

Achievements

- State-level Bronze Winner in KickBoxing
- Gold Winner in Ju Srijan (Tech Fest of Jadavpur University)

OBJECTIVE

M.Tech student in Intelligent Automation & Robotics with strong expertise in Computer Vision, Robotics, Full-Stack Development, and Interactive Simulations. Seeking AI/ML, Robotics, and Software Engineering roles where I can apply my research and development skills.

PROFESSIONAL EXPERIENCE

**Programmer Analyst Intern(10/23/2023 to 12/27/2023)**, Cognizant  
*Offer Letter View   Resignation Letter View*

- Worked on enterprise-level software development projects as part of the Programmer Analyst team.
- Gained hands-on experience with industry-standard development practices and agile methodologies.
- Contributed to full-stack development tasks, collaborating with cross-functional teams.
- Participated in code reviews, debugging, and testing to ensure high-quality software delivery.

EDUCATION

|              |                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------|
| 2024–Present | <b>M.Tech (Intelligent Automation and Robotics)</b> , Jadavpur University<br>CGPA: 8.47/10, YGP: 77.75% |
| 2019–2023    | <b>B.Tech (Electronics and Communication)</b> , St. Thomas College<br>CGPA: 8.33/10                     |
| 2017–2019    | <b>Patha Bhavan</b> (12th: 65%)                                                                         |
| 2015–2017    | <b>Patha Bhavan</b> (10th: 83.28%)                                                                      |

SELECTED CERTIFICATIONS

- Fullstack Development - Coding Ninjas
- Python (Basic) - HackerRank
- SQL (Basic) - HackerRank
- BSNL Telecom Training

TECHNICAL ASSESSMENTS

- **TCS iON National Qualifier Test (NQT) 2026**
  - Cognitive Ability: 82.85% | Advanced Quantitative & Reasoning: 93.97%
  - Programming (Python): 72.76%
  - Official Scorecard

PROJECTS

Full-Stack & Mobile Development

- **Kanban Connection Board** — MERN Stack, React, Node.js, Express.js, MongoDB, JWT, Hugging Face API, Three.js  
**Live Demo   GitHub   Admin Portal   Admin GitHub** (*Backend available on request*)
  - Developed a full-stack Kanban productivity platform featuring secure user authentication, task management, and interconnected task visualization via a Connection Board.

- Implemented AI-powered chatbot assistance and intelligent task suggestions leveraging the Hugging Face API.
- Added drag-and-drop task management for intuitive movement of tasks across workflow stages.
- Built secure JWT-based authentication with persistent task storage and automatic task creation timestamps.
- Developed an administrative dashboard for monitoring user activity, reviewing tasks, tracking creation times, and flagging inappropriate content.
- Enhanced user experience with responsive UI designs and Three.js visual elements.
- Designed collaborative board functionalities to enable multi-user teamwork.

• **Android Job Posting Platform** — Java, Kotlin, MERN Stack

**Showcase**

- Built a full-stack job platform with secure REST APIs using Node.js, Express, and MongoDB.
- Integrated CSV bulk upload functionality for efficient job posting management.
- Developed a responsive Android UI with seamless client-server integration.

## Game Development & Simulation

• **DexterGallery** — React, Three.js, WebGL

**Live Demo** **GitHub**

- Built an immersive 3D interactive simulation platform for exploring diverse scientific domains, including Physics, Chemistry, Astronomy, Robotics, and Quantum Mechanics.
- Developed a Discovery Hub with real-time 3D lab simulations, enabling hands-on virtual experimentation across multiple disciplines.
- Designed modular lab environments (Radiation Physics, Optical Deflection, Semiconductor, Electromagnetism, Kinematics, Mathematics, and Computer Science) for educational and research purposes.
- Designed real time 3D IoT labs interactive with the environment using OpenCV Python under IOTHub.
- Implemented a React-based frontend with seamless navigation and responsive UI for an intuitive user experience.

• **Industrial Multi-Agent Robot Navigation Simulator** — Unity 6, C#

**Demo Video**

- Developed a server-authoritative multiplayer simulator featuring A\*, Dijkstra, RRT, and NavMesh navigation algorithms.
- Integrated support for Python-trained PyTorch models to execute custom navigation behaviors.
- Implemented full multiplayer synchronization utilizing Netcode for GameObjects.

• **Inception Battlegrounds** — Roblox Studio, Lua

**Play**

- Designed a multiplayer combat experience with custom gameplay systems, progression mechanics, and UI.
- Executed full lifecycle development from initial design to final deployment.

## Robotics & AI

• **CRMSF: Vision-Based Gesture-Controlled Robotic Arm** — Python, TensorFlow, MediaPipe, YOLO, ROS, Gazebo

**Demo**

- Developed a real-time vision-based robotic teleoperation system for intuitive control of a 6-DOF robotic arm and dexterous gripper using monocular hand tracking.
- Designed the **CNN Regressive Motion Stabilization Filter (CRMSF)** to reduce hand-tracking jitter and improve the stability and smoothness of robotic motion.
- Integrated MediaPipe hand tracking, YOLO-based object perception, and ThreeJS WebGL simulation into an end-to-end gesture-to-actuation pipeline for low-latency robotic control.
- Research manuscript currently under preparation.

• **MobiUNet-3D: Hybrid Brain Tumour Segmentation and Volumetric Reconstruction** — TensorFlow, Keras, Python

**Live Demo**

- Developed a lightweight hybrid deep learning framework with a shared MobileNetV2 encoder for simultaneous brain tumour classification and MRI segmentation, reducing redundant computation through multi-task learning.

- Designed a unified segmentation-classification architecture with dedicated task-specific heads, enabling efficient feature reuse while maintaining a compact model footprint.
- Implemented a volumetric reconstruction pipeline that converts segmented MRI slices into patient-specific STL models using the Marching Cubes algorithm for three-dimensional visualisation.
- Integrated preprocessing, data augmentation, and K-fold cross-validation for robust model evaluation. Research manuscript under review.

• **Robotics Car Project**

**Drive**

- Implemented IR sensor-based obstacle detection with real-time data streaming to backend.
- Created a Godot Engine interface with Python for data visualization and processing.

## BACHELORS FINAL YEAR PROJECT

• **Compact Wideband MIMO Antenna for 5G and IoT**

**Report**

- Designed and simulated a compact four-element octagonal microstrip MIMO antenna using **CST Studio Suite**.
- Achieved **2.64–20 GHz** bandwidth, **>20 dB isolation**, **6.53 dB peak gain**, and **ECC < 0.01**.
- Fabricated and validated prototype in an anechoic chamber.

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